MINIFLY V193 / EXPERIMENT DESIGN / 01

Reuse a learned relation after the world changes

Primary question. After learning parity in one world, can MiniFly update its structural maps and answer untaught pairs correctly after a graph change, with its relational readout frozen and zero new relation labels?

Why this is the next test

In V191, direct online updates reached 98.44% on both changed and still-valid pairs after 32 post-change events. Slow deployment reached 78.65% and 95.31%, respectively. The direct arm averaged 99.48% still-valid accuracy across checkpoints, leaving only 0.52 points of headroom against a five-point promotion requirement. Keep the original verdict and use direct online as the reference.

V191 updated the readout in every arm. Its 32-event result also followed 512 observed background symbols, 896 map-row updates and 32 relation labels. V193 measures structural experience and relation feedback separately.

Common starting state

Reuse the V191 byte interface, one-step and two-step local TD maps, fixed sparse features, local parity readout and learned newline table. For each world, learn one common state through the clean and noisy pre-change phases. Clone that state at the graph change, including maps, readout, newline table and counters. Keep the query parser and answer-byte mapping fixed. Apply the same adjacent opposite-parity swap as V191. Keep swap identity and phase boundaries evaluator-only.

Four paired arms

A. Joint reference. Update maps directly and update the readout on permitted post-change labels.

B. Reusable operator. Update maps directly. Clamp the readout and deliver zero post-change relation labels.

C. Frozen maps. Clamp maps and continue teaching the readout. This measures compensation by new labels.

D. Fully frozen. Clamp maps and readout and deliver zero new relation labels. This measures persistence of the old predictions.

All arms observe the identical background stream and receive identical query prefixes at scheduled teaching and probe slots. The feedback channel alone carries answer bytes, and it is absent for B and D. Keep query records and feedback outside map updates. Freeze the learned newline table in all arms during the post-change phase. No replay, backpropagation, attention, or content-addressed reads.

Measure transfer without reteaching

Sealed pairs and checkpoints

Use 12 fresh paired worlds for the primary clean post-change test. Seal a fixed, label-balanced panel of 16 ordered changed pairs and 16 ordered still-valid pairs per world. Hold out their reversals as well. Verify that every scored pair and reverse remains absent from relation feedback throughout the lifetime. Supply clean post-change background in the primary suite. Run continued 10% background corruption on a separate, sealed secondary roster, scored independently.

Probe before any post-change exposure and after 8, 32, 128 and 512 background blocks. Each block has 16 background bytes, 14 sampled updates per stride in map-learning arms, and one scheduled teaching query. At the 32-block checkpoint, B has seen 512 background bytes and made 896 map-row updates with zero relation labels. A has also received 32 labels. Report actual counts by arm and checkpoint.

Scoring and controls

Score the first generated answer plus the model-generated newline with writes disabled. Report answer-only accuracy, both panel denominators, and the fraction of changed pairs still answered with their obsolete label. Record predictions before any feedback. Audit map and readout hashes separately and verify B and D receive no label bytes in learner state or callbacks.

Keep the observed-transition graph comparator. Fit it from the same background observations with a predeclared observation window and infer polarity from pre-change feedback only for its zero-label run. Its equal-group prior makes it a task reference. Include exact and unordered pair lookup and a constant-answer baseline. Randomize byte identities and answer polarity by world.

Proposed primary gate

At 128 blocks, require B to reach at least 85% joint accuracy on changed pairs and 90% on still-valid pairs. Require at least a 10-point paired mean advantage over D on changed pairs, with a paired 95% interval excluding zero. Report the gap to A as the cost of withholding labels. Score 512 blocks as a durability and late-recovery checkpoint. Report all checkpoints even if the gate fails.

Before interpreting B, verify A reproduces at least 85% changed-pair accuracy and 90% still-valid accuracy at 128 blocks. If A misses, treat the lifetime as an invalid reference comparison and audit the interface and exposures. Keep the noisy suite exploratory and report its results separately.

Decisions and follow-up

Interpretation

B passes. Use the direct structural learner with a reusable readout as the reference. The result supports a narrow E3 claim: on this synthetic parity task, structural observations alone enable a previously taught operator to answer untaught pairs after a world change. The fixed parser supplies operand roles.

A passes and B fails. The current map and readout need joint adaptation at this dose. Inspect whether post-change map features preserve the learned decision geometry before trying an invariant interface or another local readout. Keep slow deployment out of the main comparison.

B passes clean and degrades under continued noise. Report the noise boundary. Use the observed-transition reference to distinguish weak structural evidence from failure to use the resulting features.

Next architecture tests

Pathway selection. After the operator test, predeclare a small bank of strides and discounts with direct updates. Compare a local feedback-driven mixture with a fixed mixture and the best single path chosen on development worlds at matched feature and readout budgets. Include an asymmetric relation and reversed operand probes.

Factual memory and integration. Keep factual storage separate from mutable structural maps. Qualify invariant encoding plus teach-gated writes and float16 storage under a measured envelope before combining it with the relational pathway. Prior float16 results cover a fixed-lag, all-byte store. Keep the explicit causal query router for the first mixed lifetime and compare integrated competence with standalone references.

Execution record

Use engineering worlds to verify branch identity, readout clamps, zero-label isolation, balanced panels, exact observation counts, and restart determinism. Freeze source hashes, seed roster, gates, comparator settings and checkpoint definitions before final worlds. Save paired scores, receipt hashes, and resource usage. Report background bytes, map-row updates, relation labels and runtime separately.

Evidence and status

Basis: user-provided audit of V191 and the published V191 results at commit 723b49ba. V193 is a design proposal. The roster, thresholds and continued-noise suite are proposed settings.